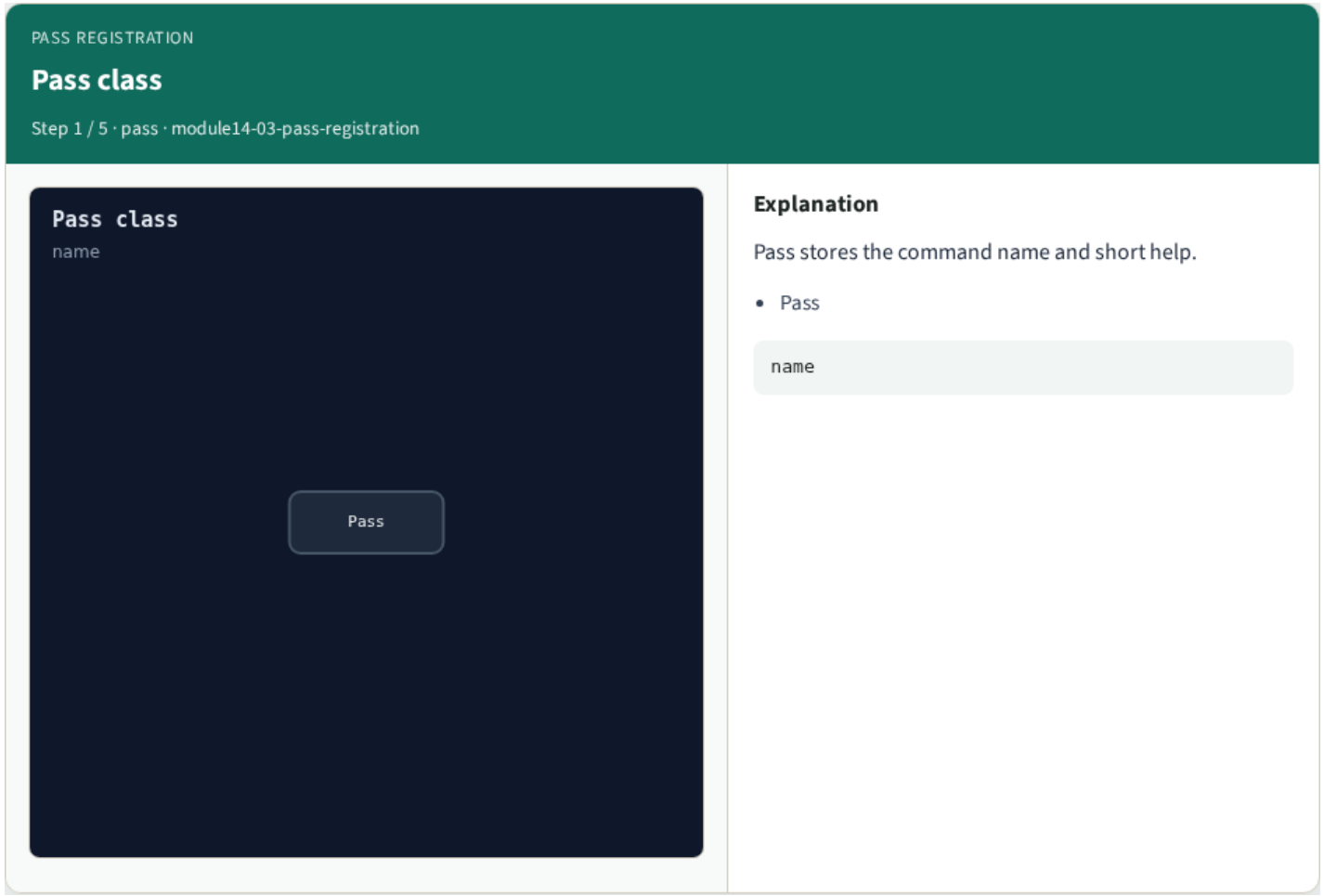
Why registration

Yosys is a pass engine

The idea

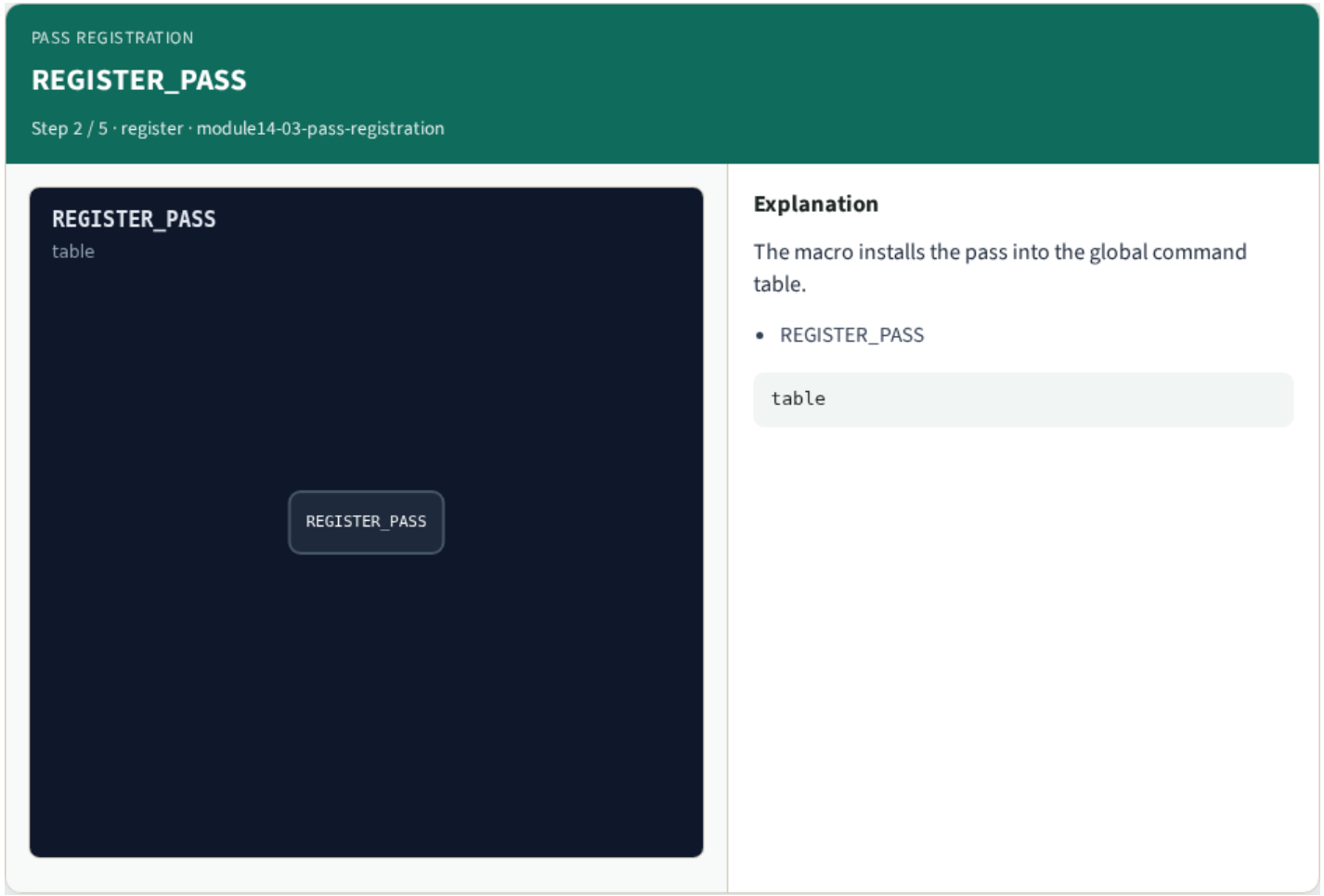
- GOLDEN: subclass Pass, REGISTER_PASS, implement execute(args, design)

Pass class



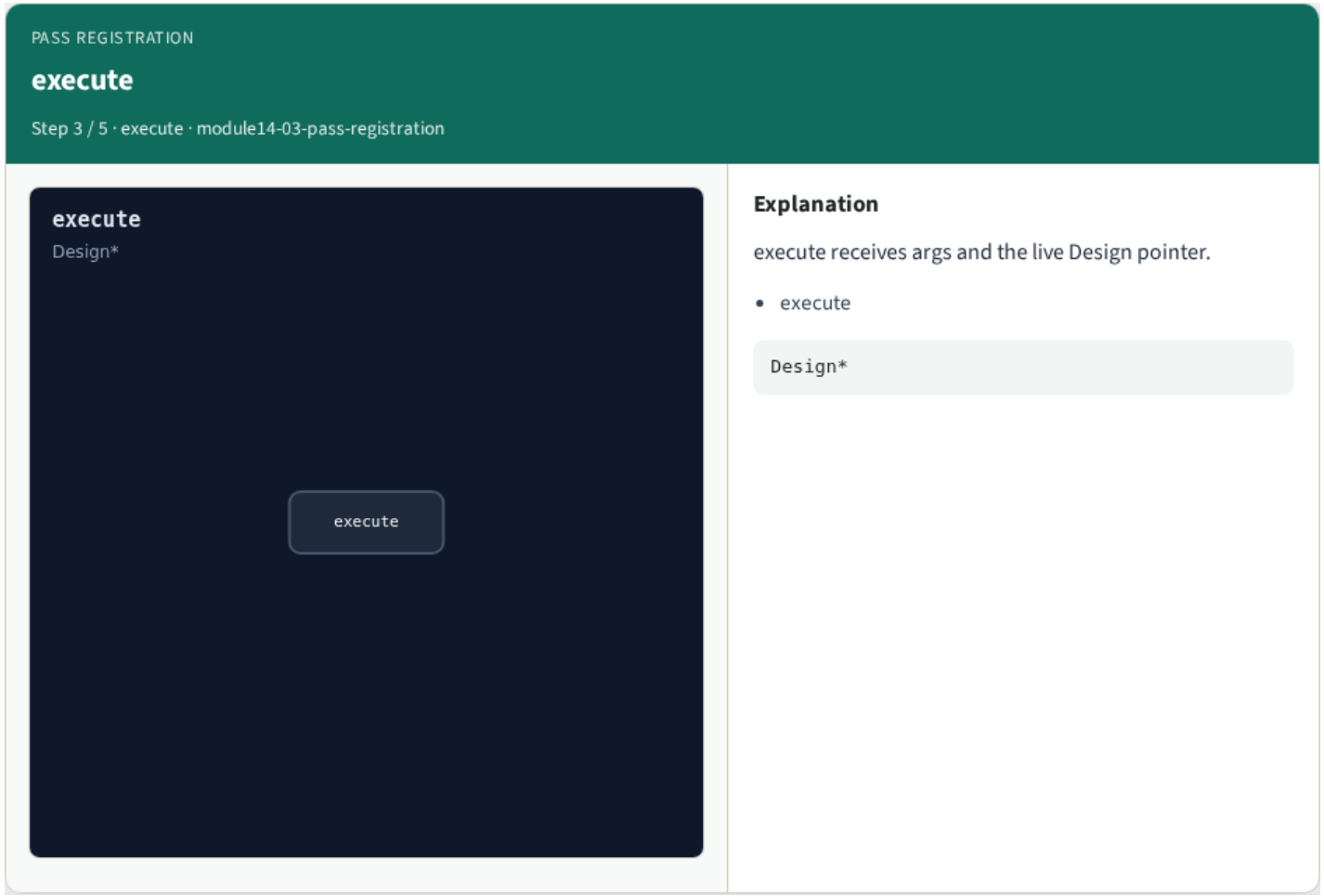
Pass class

REGISTER_PASS



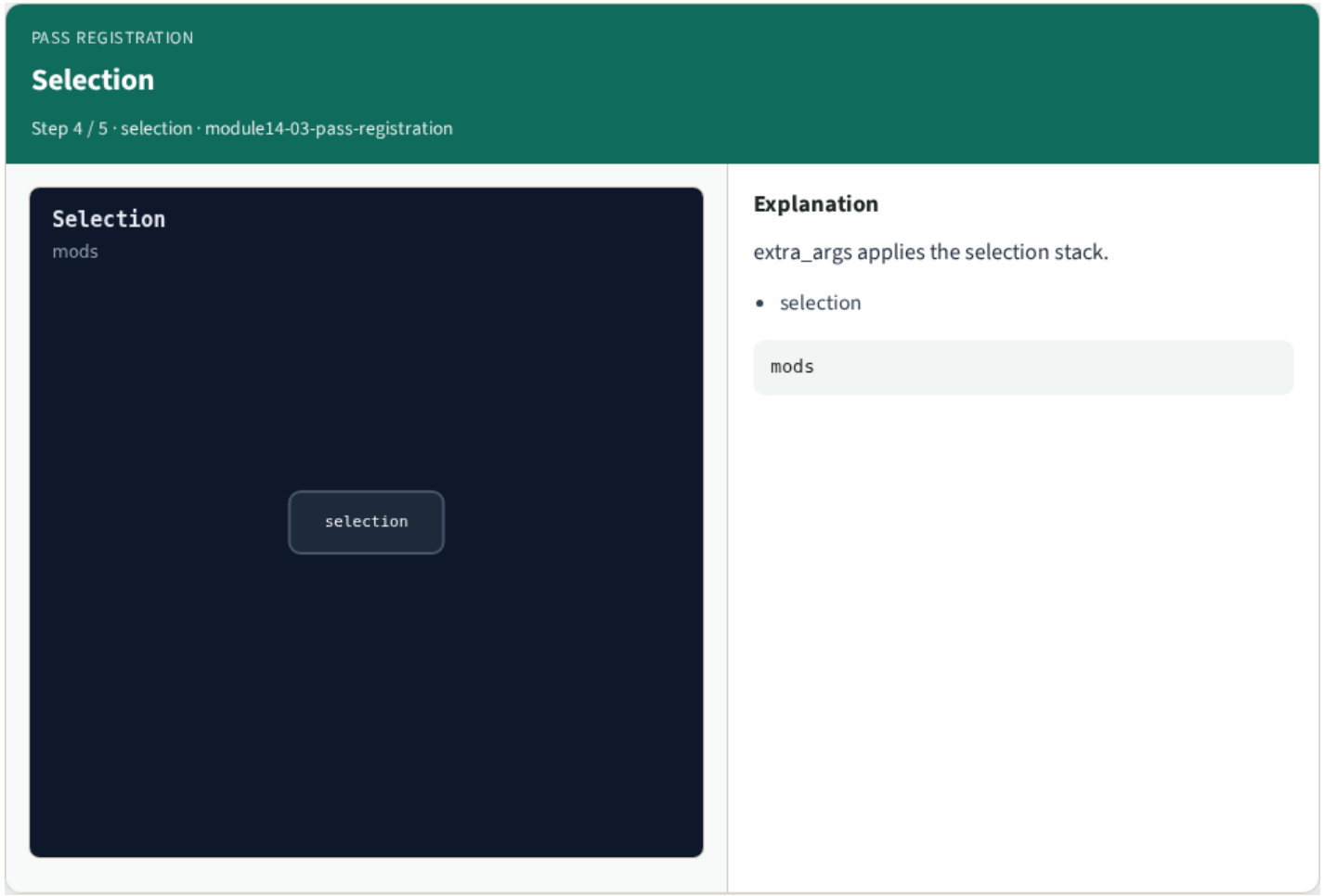
REGISTER_PASS

execute



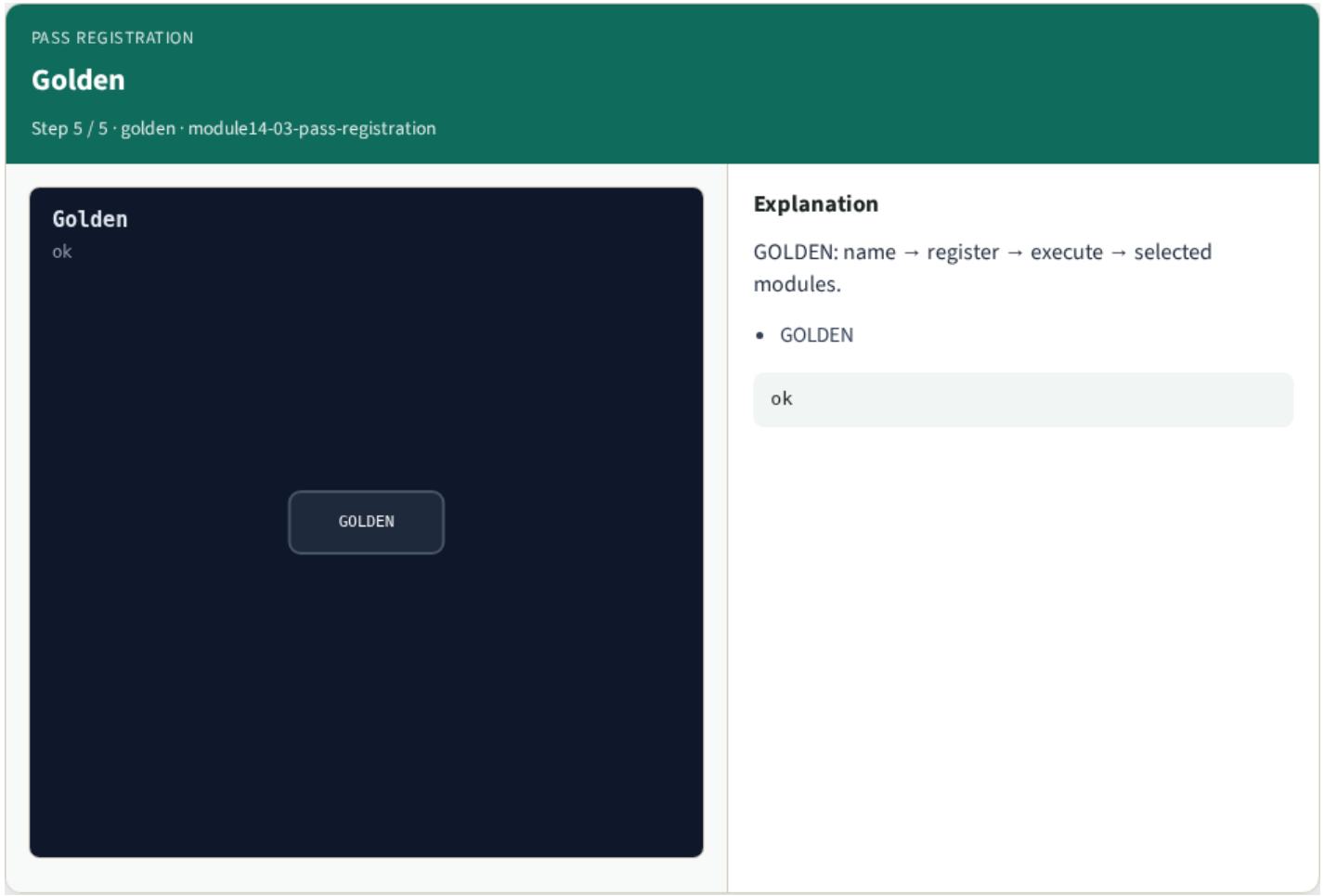
execute

Selection



Selection

Golden



Golden

Browser lab track

- Open **pass-registration**
- Order the registration steps and check

Implement track

- Call pass_registration(); assert recipe ids match the golden sequence

Pitfalls

- Forgetting REGISTER_PASS
- Mutating modules that are not selected

Try these

- From the common folder, call pass_registration and print the recipe

Try these — try these

```
cd courses/learn_yosys/common
python3 -c "from internals_solvers import pass_registration;
print(pass_registration())"
```

Pseudocode

- Here is the Pass registration shape in plain steps

Pseudocode — try these

```
struct MyPass : Pass { MyPass(): Pass("my_pass", "help") {}
  void execute(args, design) override { ... } };
REGISTER_PASS(MyPass)
```

Next

- Complete the quiz for this part
- Next: read the real opt_clean shape in the Yosys tree